Rosario Cecilio-Flores-Elie

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Research Interests

Planetary Surface Processes | Icy Ocean Worlds (Titan, Enceladus) | Geomorphological Mapping | Structural Geology | Near-Surface Geophysics | Terrestrial Analogs

Education

M.S. in Geological Sciences – Cornell University 2026
M.S. in Astrophysics – CUNY Graduate Center 2024
B.S. in Physics – CUNY Lehman College 2022
M.S.Ed. in Literacy Education – Alfred University 2015
B.A. in Sociology – CUNY Lehman College 2013

Research Experience: Planetary Science and Astrophysics

NASA Dragonfly Guest Investigator – Johns Hopkins Applied Physics Lab Nov. 2024 – Dec. 2026
Mentor: Emily S. Martin *Laurel, MD / Hybrid*
Project: Investigating Regional Controls on Titan's Polygonal Impact Craters

- Mapped the surface features and fractures around 15 polygonal impact craters on Titan from Cassini SAR imagery in ArcGIS Pro.
- Built a catalog of these features, including coordinates, length, and azimuth (fractures), to support future modeling of why some craters do not fit the global stress pattern.

Graduate Research Fellow – American Museum of Natural History Aug. 2022 – Sept. 2024
Advisor: Jacqueline Faherty *New York, NY*
Thesis: From Stars to Moons: Investigating Stellar Rotations, Planetary Interactions, and Exoplanetary Prospects

- Compiled and analyzed physical properties of planet-moon systems to identify patterns among geologically active ocean worlds such as Europa, Titan, and Enceladus.
- Processed and analyzed 99 NASA TESS light curves in Python (Lomb–Scargle periodograms) to measure stellar rotation periods, contributing to a peer-reviewed publication (Rothermich et al. 2024).

Research Experience: Geology and Near-Surface Geophysics

Independent Research: Karst Analogs for Titan Sept. 2026 – Present
Project: Cenotes as Fracture Proxies in the Yucatán Peninsula, México

- Building an initial catalog of karst features (cenotes) in the Yucatán Peninsula using Copernicus 30m DEM and Landsat imagery in ArcGIS Pro to better understand fracture-controlled fluid pathways on Earth and icy ocean worlds.

M.S. Researcher – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Aug. 2026
Advisor: Carolina Muñoz-Sáez *Ithaca, NY*
Thesis: Secondary Structural Lineaments and Hydrothermal Manifestation Distribution at El Tatio Geothermal Field

- Investigated how fault and fracture networks control where hydrothermal fluids reach the surface using drone-derived optical and thermal orthomosaics and DEMs (Structure from Motion, Agisoft Metashape) in ArcGIS Pro.
- Mapped surface temperature at more than 1500 hydrothermal features across two basins and used Python statistics (Mann–Whitney U, Spearman) to test vent temperature against distance to faults.

NSF REU: Water Quality in the Yucatán Peninsula – Northern Illinois University June – Aug. 2021
Mentors: Philip J. Carpenter & Melissa Lenczewski *Quintana Roo, México*
Project: Karst Conduit Identification Using Geophysical Surveys in Northeast Yucatán, México

- Conducted electrical resistivity surveys with a SuperSting™ system to image fractures, saturated zones, and karst conduits near cenotes along the Holbox Fracture Zone.

- Identified previously unmapped subsurface features in the town of Agua Azul, including an air-filled cavity and a water-saturated conduit about 5m below the surface, consistent with fracture-controlled groundwater flow.

Technical Skills

Remote Sensing: Cassini synthetic aperture radar (SAR) imagery interpretation; thermal infrared and radar image interpretation; planetary image processing with USGS ISIS

GIS & Geospatial Analysis: ArcGIS Pro (intermediate), QGIS (intermediate): structural lineament and regional geomorphologic mapping, proximity and spatial statistics with Python, DEM analysis

Drone Photogrammetry: Structure from Motion (SfM) processing in Agisoft Metashape; optical (RGB) and thermal infrared (TIR) orthomosaic generation; DEM production and analysis

Programming & Data Analysis: Python (intermediate): statistical analysis (Mann–Whitney U, Spearman correlation), time series analysis (Lomb–Scargle periodograms), large dataset processing; supervised machine learning (beginner)

Field & Geophysical Methods: Near-surface geophysical surveys in karst terrain; electrical resistivity tomography for imaging fractures and saturated zones, SuperSting™ system (intermediate)

Writing: LaTeX / Overleaf

Languages: Bilingual and biliterate in English and Spanish

Publications

Theses

Cecilio-Flores-Elie, Rosario (2026). "Secondary Structural Lineaments and Hydrothermal Manifestation Distribution at El Tatio Geothermal Field". M.S. thesis. Cornell University. URL: https://www.proquest.com/dissertations-theses/secondary-structural-lineaments-hydrothermal/docview/3385816715/se-2?accountid=10267.

Cecilio-Flores-Elie, Rosario (2024). "From Stars to Moons: Investigating Stellar Rotations, Planetary Interactions, and Exoplanetary Prospects". M.S. thesis. CUNY Graduate Center. URL: https://academicworks.cuny.edu/gc_etds/6013.

Peer-Reviewed, Co-Author

Rothermich, Austin et al. (2024). "89 New Ultracool Dwarf Comoving Companions Identified with the Backyard Worlds: Planet 9 Citizen Science Project". In: *The Astronomical Journal* 167.6, p. 253. DOI: 10.3847/1538-3881/ad324e.

Conferences and Presentations

"Clues in the Crust: Titan's Polygonal Impact Craters and Their Regional Signals" Nov. 2026
Oral, Dragonfly Annual Team Meeting, Laurel, MD

"Where Faults Lead, Fluids Follow: Inferring Structural Controls on Hydrothermal Activity for Ocean World Exploration" Sept. 2026
Poster, AbGradCon, Tucson, AZ

"Fault-Controlled Fluid Pathways and Geyser Expression in the El Tatio Geyser Field: Implications for Plume Activity on Icy Ocean Worlds" May 2026
Poster, Astrobiology Science Conference (AbSciCon), Madison, WI

"Exploring Mass Ratios in Planetary–Moon Systems: Insights from Our Solar System and Beyond" Jan. 2024
Poster, American Astronomical Society (AAS), New Orleans, LA

"Mass Ratios of Planets and Active Moons: Insights for Ocean World Observation" Dec. 2023
Poster, American Geophysical Union (AGU), San Francisco, CA

"Ocean Worlds: To Enceladus and Beyond" (Public-Facing Translation) June 2023
Poster, CUNY Science Communication Symposium, New York, NY

"Karst Conduit Identification Using Geophysical Surveys in Northern Yucatán, México" Oct. 2022
Poster, SACNAS NDiSTEM Conference, San Juan, PR

"Karst Conduit Identification Using Geophysical Surveys in Northern Yucatán, México" Dec. 2021
Oral, American Geophysical Union (AGU), New Orleans, LA

Fellowships and Awards

Travel Grant, Space Imagery Center Planetary Photogrammetry Workshop Dec. 2026

Long Fellowship, Dept. of Earth and Atmospheric Sciences, Cornell University 2024 – 2025

Best Poster Award, CUNY Science Communication Symposium June 2023
Public-Facing Translation category

Student Travel Grant, American Geophysical Union (AGU) Dec. 2021

Teaching and Mentoring

Teaching Assistant, EAS 2250: Earth Systems (Lab) Aug. 2025 – May 2026
Dept. of Earth and Atmospheric Sciences, Cornell University *Ithaca, NY*

- Led weekly lab sections for an introductory Earth systems course covering geology, hydrology, oceanography, and atmospheric science.
- Designed short interactive lessons before each lab and gave written feedback on lab reports to strengthen students' knowledge and interpretation.

Second Grade Bilingual Teacher & Curriculum Lead Aug. 2013 – June 2022
P.S./I.S. 218 Rafael Hernandez Dual Language Magnet School *Bronx, NY*

- Designed and taught dual-language (English and Spanish) curricula for 25–50 students each year.
- Led grade-level literacy curriculum design aligned with state and federal standards.
- Mentored 1–2 student teachers each year through US PREP and CUNY Lehman College.

Outreach and Science Communication

Letter to a Pre-Scientist, Science Pen Pal Program Sept. 2024 – June 2026

Spanish Translator Apr. 2023 – Sept. 2024
Machine Learning for Physics and Astronomy (Flatiron Institute Open Learning)

Public Talks and Programs
Open Space – Planetarium Show (Co-ran) May 2024
Hunter High School Field Trip, American Museum of Natural History, New York, NY

Decoding the Night Sky: Exploring Maya Astronomy April 2023
Science StoryTellers, Variety Boys and Girls Club of Queens, Queens, NY

Professional Memberships

American Geophysical Union (AGU)
Geological Society of America (GSA)
American Astronomical Society (AAS)
Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)